

Low Level Design (LLD) - RAG Customer Support System

1. Module Design

Document Processing Module: PDF ingestion.
Chunking Module: Splits documents.
Embedding Module: Converts text to vectors.
Vector Storage Module: Stores embeddings.
Retrieval Module: Finds relevant chunks.
Graph Module: Controls workflow.
HITL Module: Handles escalation.

2. Data Structures

Document: Raw PDF text.
Chunk: {text, metadata}.
Embedding: vector representation.
Query Schema: user input string.
Graph State: {query, context, response, status}.

3. Workflow Design

Nodes: Retrieve → Process → Output.
Edges: Conditional transitions.
State: Shared memory across nodes.

4. Routing Logic

If context found → generate answer.
If missing context → escalate.
If low confidence → HITL trigger.

5. HITL Design

Triggered on ambiguity.
Human provides clarification.
Response reintegrated into flow.

6. API Design

Input: {query}.
Output: {answer, confidence, source}.
CLI-based interaction supported.

7. Error Handling

No chunks → escalate.
LLM failure → retry or fallback.
Invalid query → prompt clarification.